Reward Shaping in Contextual Bandits for E-commerce

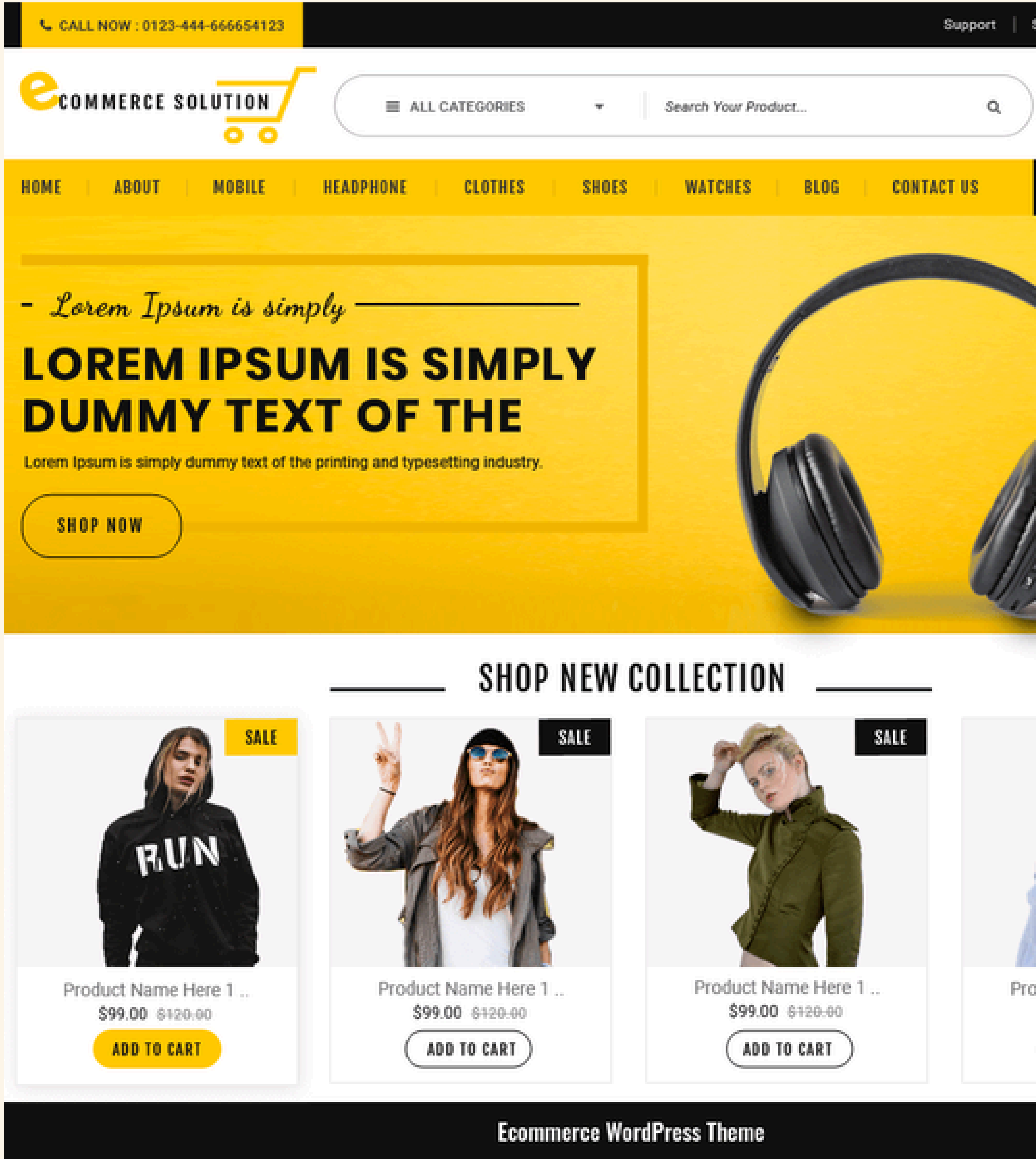
Presented by

Eden Amram

Ziv Shamli

Motivation

- Goal: maximize purchases in e-commerce
- Purchases are valuable but sparse
- Earlier funnel actions may indicate buying intent
- Reward design affects what the bandit learns



E-COMMERCE AS A FUNNEL PROCESS

- E-commerce behavior follows a funnel structure
- Users usually move from early actions to deeper actions
- Deeper stages indicate stronger purchase intent
- Purchase is the final and most valuable stage

CONTEXTUAL BANDITS

Basic Idea

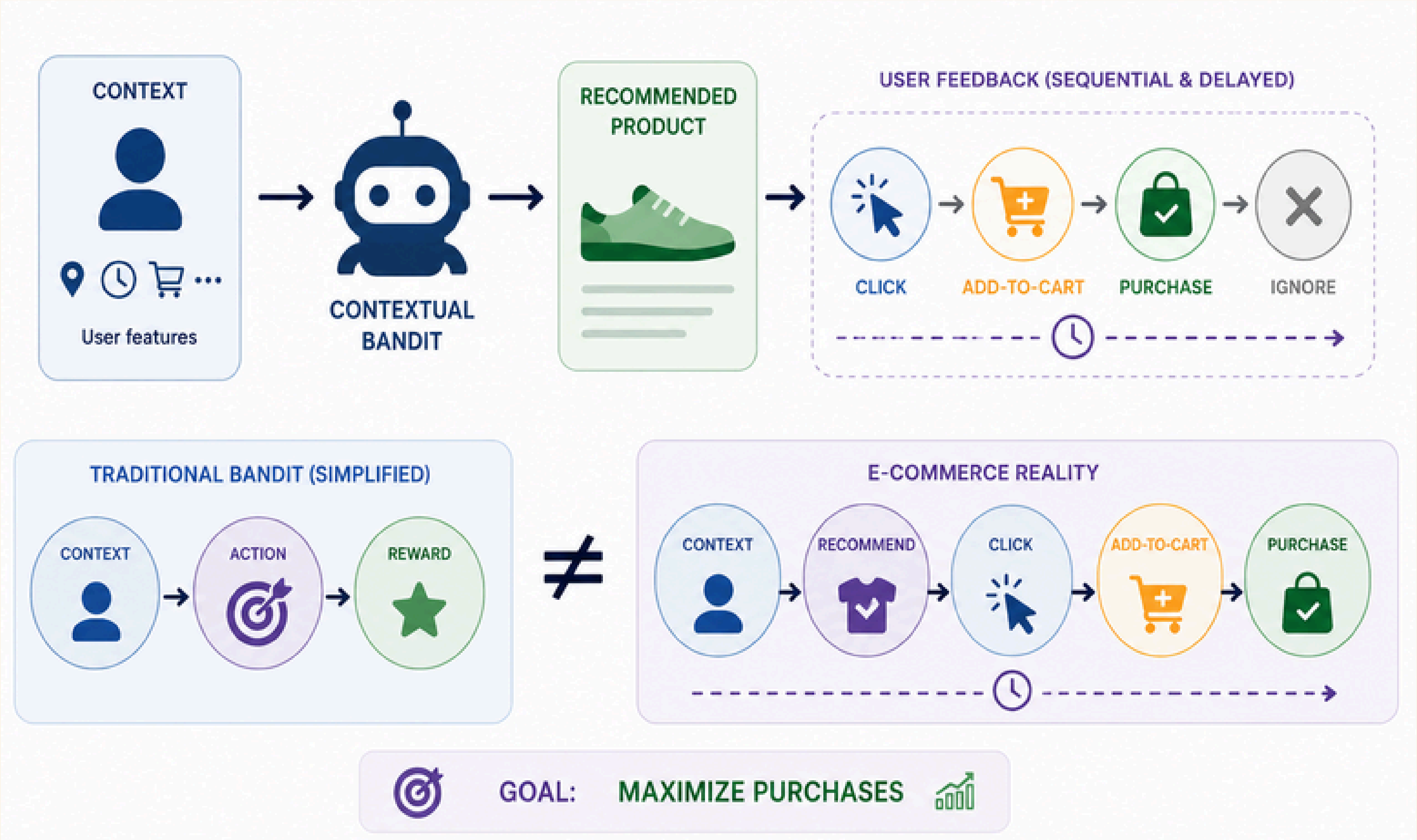
A contextual bandit selects an action based on the current context and learns from user feedback.

In E-commerce

- Context → user information
- Action → recommended product
- Reward → user interaction

Why useful here?

- Learns from real user behavior
- Personalizes recommendations
- Balances exploration and exploitation



Binary Reward - Click / No-click Feedback

Semenov et al. - "Diversity in News Recommendations Using Contextual Bandits"

Reward definition:

Click = 1
No click = 0
Each article is treated as an arm.

Main issue:

Optimizing only clicks may over-expose popular articles and reduce recommendation diversity.

Their solution:

DivLinUCB adds a cost to repeatedly recommended articles.
The goal is to maintain a balance between clicks and diversity.

This formula calculates a score for each article, and the algorithm selects the article with the highest score.

$$D_{i,a} = \underbrace{\frac{S_a^T f_{i,a}}{1+\beta\left(\frac{rc_a}{\hat{T}}\right)^2}}_{\text{Exploitation adjusted by diversity}} + \underbrace{\alpha\sqrt{f_{i,a}^T X_a^{-1} f_{i,a}}}_{\text{Exploration (UCB bonus)}}$$

Weighted Reward

Dragone et al. - "Deriving User- and Content-specific Rewards for Contextual Bandits"

Baseline reward:

Stream ≥ 30 seconds = 1
Otherwise = 0

Main issue:

A fixed binary reward ignores different levels of user engagement.

Their solution:

short stream → low reward
medium stream → medium reward
long stream → high reward

Key takeaway:

Weighted rewards provide richer feedback than a simple binary signal.

Max Stage Reward - Deepest Funnel Stage

Xu et al. — Decision Making Problems with Funnel Structure

Paper context:

Contextual bandit with funnel structure in email marketing campaigns.

Funnel stages:

Open → Click → Purchase

Main idea:

Purchase is the most valuable signal, but it is very sparse.

Their approach:

Use information from all funnel stages, not only the final purchase signal.

Research Question

WHICH REWARD DEFINITION HELPS A CONTEXTUAL BANDIT MAXIMIZE PURCHASES?

Missing gap:

No controlled comparison between these reward definitions.

Binary Reward

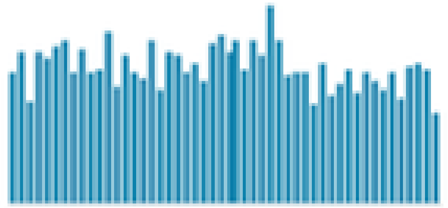
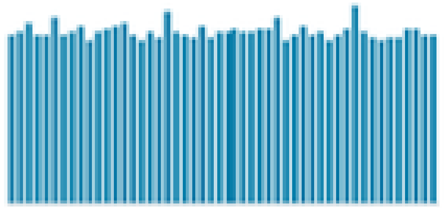
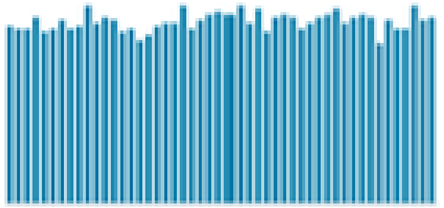
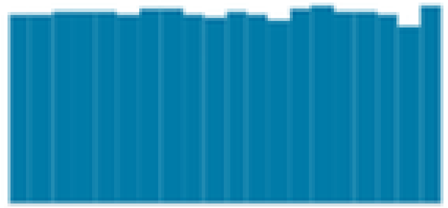
Weighted Reward

Max Stage Reward

Research Methodology

- Investigated the impact of reward shaping on contextual bandit recommendation systems.
- Compared three reward formulations under identical experimental conditions.
- Used the same dataset, algorithms, and evaluation protocol.
- The reward function was the only variable changed between experiments.
- This enabled a fair and controlled comparison of reward strategies.

Retailrocket Recommender System Dataset

# timestamp	visitorid	event	itemid	transactionid
the time, when event is occurred, in milliseconds since 01-01-1970 (Standard Unix timestamp	unique identifier of the visitor	type of the event {"view", "addtocart", "transaction"}	unique identifier of the item	unique identifier of the transaction (non empty only for transaction event type).
 1431b 1443b	 0 1.41m	view 97% addtocart 3% Other (22457) 1%	 3 467k	 0 17.7k

categoryid	parentid
1016	213
809	169
570	9
1691	885
536	1691
231	
542	378
1146	542

# timestamp	visitorid	event	itemid	transactio...
1433223697356	598426	view	156489	
1433224078165	223343	view	402625	
1433222531378	57036	view	334662	
1433223239808	1377281	view	251467	
1433223236124	287857	addtocart	5206	
1433224244282	1370216	view	176721	
1433221078505	158090	addtocart	10572	

# timestamp	itemid	property	value
1433041200000	183478	561	769062
1439694000000	132256	976	n26.400 1135780
1435460400000	420307	921	1149317 1257525
1431831600000	403324	917	1204143
1435460400000	230701	521	769062
1433041200000	286407	202	820407
1438484400000	256368	888	437265 1296497 n24.000 229949 651738 285933

Reward Functions

1. R1: Binary Reward : Purchase / No Purchase

2. R2: Weighted Reward
   w1*View+ w2*Cart + w3* Purchase

3. R3: Max-Stage Reward - Uses the deepest action reached in the conversion funnel.

Goal: Compare simple, weighted, and funnel-based reward strategies.

Experimental Setup

	R1	R2	R3
LinUCB	✓	✓	✓
DivLinUCB	✓	✓	✓
Random	✓	✓	✓

- All experiments were performed using the Retailrocket Dataset.

Evaluation Protocol

- **Warm-up phase (50%) :**
  - Algorithms learned from historical interaction logs.

- **Evaluation phase (50%):**
  - Algorithms made independent recommendations.

- Performance was computed using historical rewards of selected items.
- Same protocol applied to all experiments.

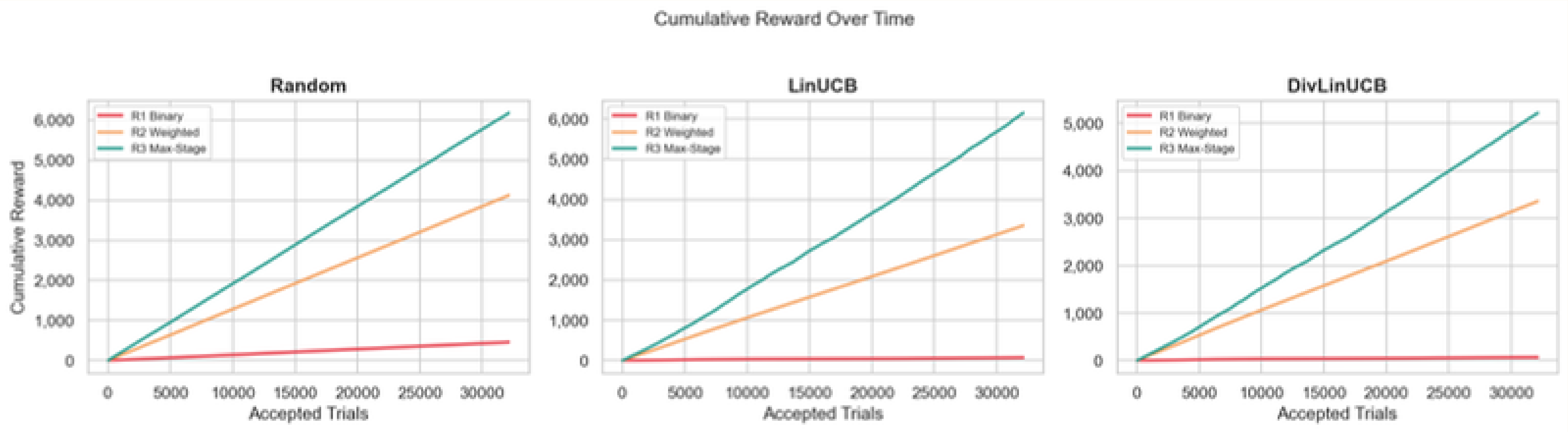
Evaluation Metrics

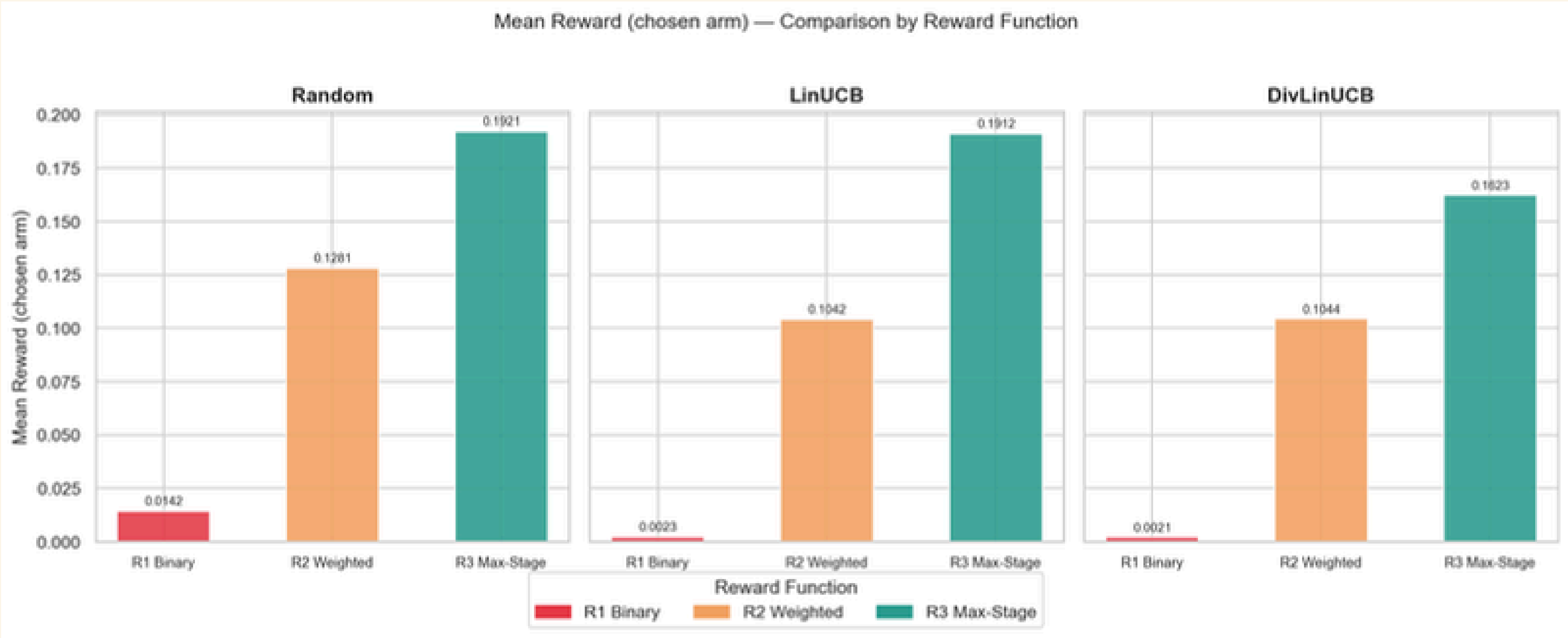
- **Mean Reward –** average reward obtained.
- **Cumulative Reward –** total reward accumulated over time.
- **Purchase Rate & CTR –** recommendation effectiveness.
- **Gini Index –** Recommendation Concentration .

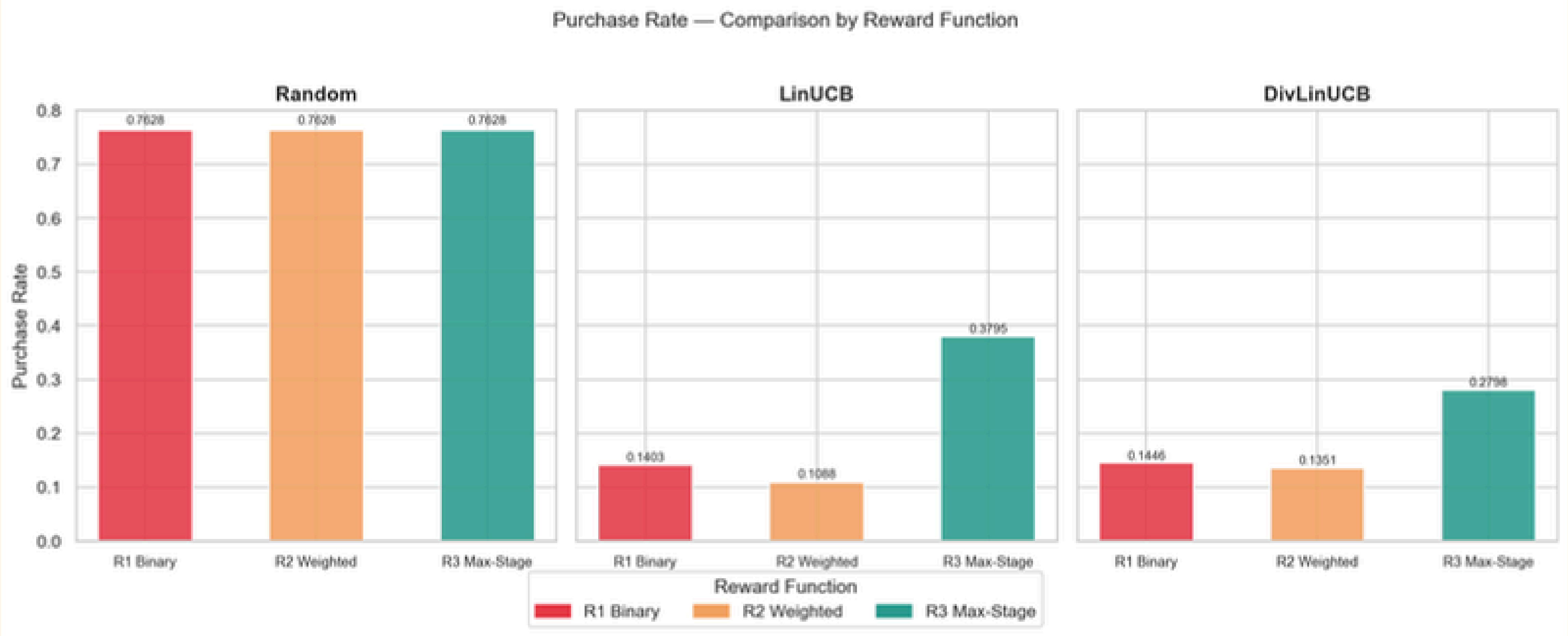
$$\text{Gini} = \frac{\sum_{a_1,a_2} |rc(a_1) - rc(a_2)|}{2|A| \sum_a rc(a)}$$

- **Learning Curves –** learning behavior .

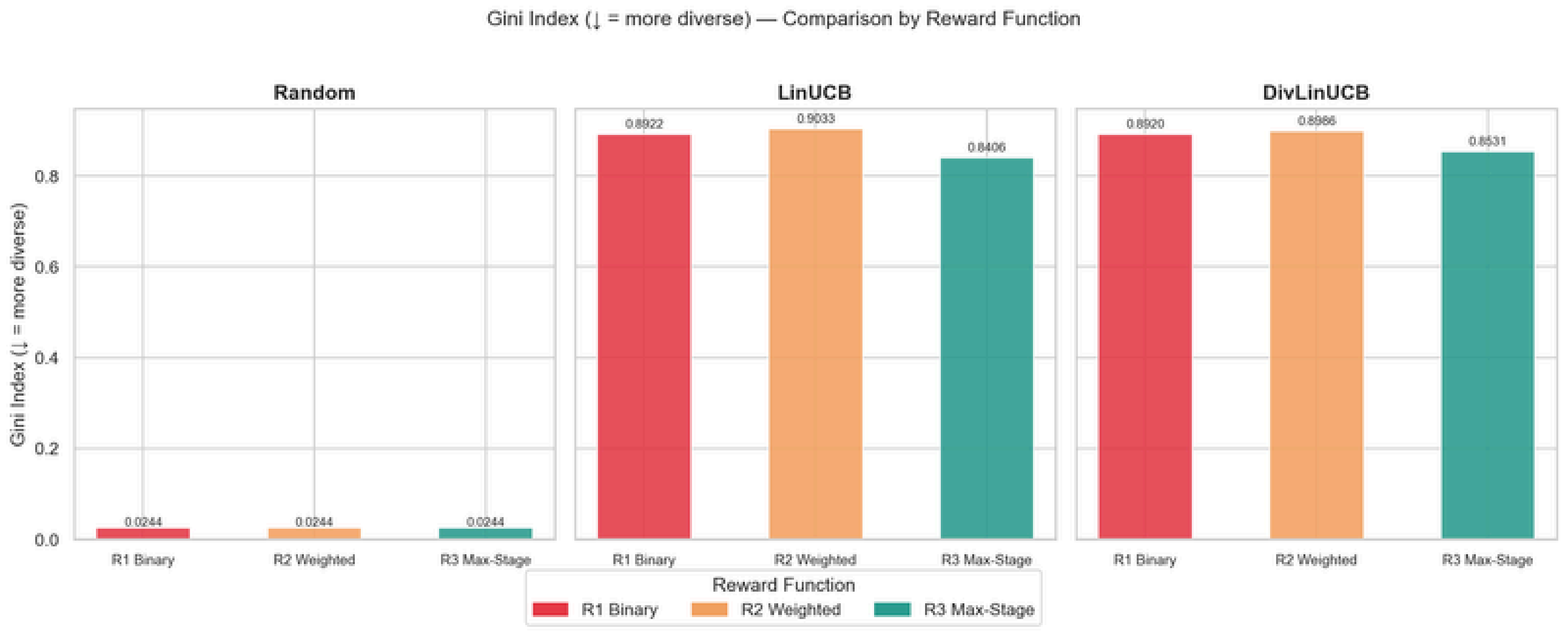
Results


Cumulative Reward Over Time
Random
LinUCB
DivLinUCB
R1 Binary
R2 Weighted
R3 Max-Stage
Cumulative Reward
Accepted Trials

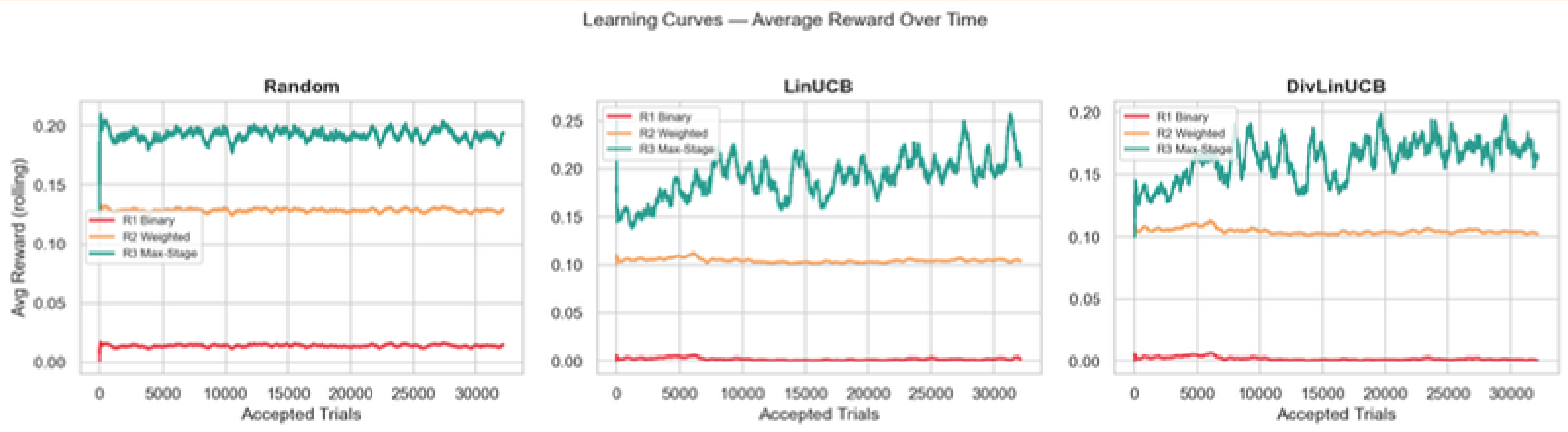

Mean Reward (chosen arm) — Comparison by Reward Function
Random
LinUCB
DivLinUCB
Mean Reward (chosen arm)
0.0142
0.1281
0.1921
0.0023
0.1042
0.1912
0.0021
0.1044
0.1623
R1 Binary
R2 Weighted
R3 Max-Stage
Reward Function



Purchase Rate — Comparison by Reward Function
Random
LinUCB
DivLinUCB
Purchase Rate
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.7628
0.7628
0.7628
0.1403
0.1068
0.3795
0.1446
0.1351
0.2798
R1 Binary
R2 Weighted
R3 Max-Stage
Reward Function
R1 Binary
R2 Weighted
R3 Max-Stage



Gini Index (↓ = more diverse) — Comparison by Reward Function
Random
LinUCB
DivLinUCB
Gini Index (↓ = more diverse)
0.0
0.2
0.4
0.6
0.8
0.0244
0.0244
0.0244
0.8922
0.9033
0.8406
0.8920
0.8986
0.8531
R1 Binary
R2 Weighted
R3 Max-Stage
Reward Function
R1 Binary
R2 Weighted
R3 Max-Stage



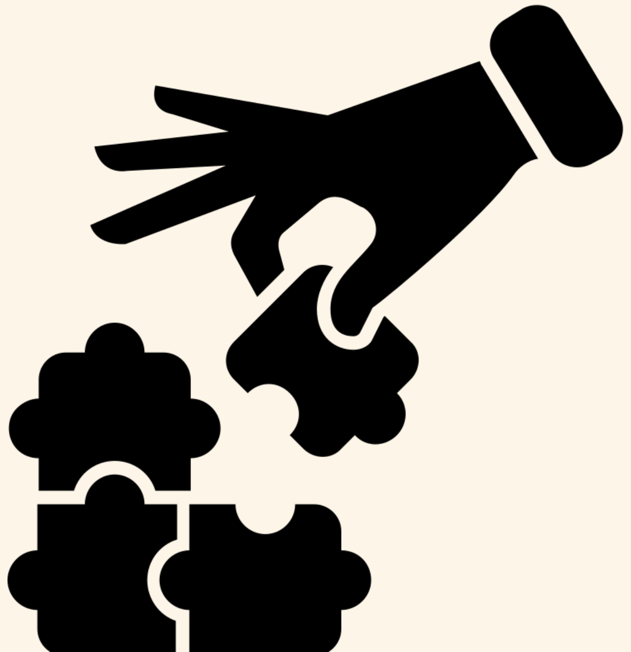
algorithm	reward	ctr
Random	R1 Binary	0.7628
Random	R2 Weighted	1
Random	R3 Max-Stage	1
LinUCB	R1 Binary	0.1403
LinUCB	R2 Weighted	1
LinUCB	R3 Max-Stage	1
DivLinUCB	R1 Binary	0.1446
DivLinUCB	R2 Weighted	1
DivLinUCB	R3 Max-Stage	1

Disscusion

- Max-Stage best reflects the sequential nature of the e-commerce conversion funnel.
- It prioritizes the deepest user interaction, providing a more focused reward signal.
- Weighted Reward incorporates multiple behaviors, but treating all stages simultaneously may reduce the importance of the final conversion event.
- Binary Reward suffers from sparse purchase feedback, limiting learning efficiency.

Conclusion

- Reward shaping significantly affects contextual bandit performance.
- Funnel-based rewards outperform traditional binary rewards.
- Max-Stage is the most effective reward strategy for e-commerce recommendations.



Future Works

- Evaluate the proposed reward strategies on additional real-world e-commerce datasets to assess generalization.
- Extend the analysis to deep reinforcement learning and advanced contextual bandit models.

Open the floor for questions and discussions.

Thank you for listening!